

Computational Wind Engineering - Project

Maccarone Giovanna s349134, Morandi Simone Domenico s336974

1. Abstract

The aim of this report is to simulate an unsteady, two-dimensional turbulent flow around a square cylinder with a Reynolds number $Re \approx 2.2 \times 10^4$, using standard $k - \omega$ model. We compare the C_p statistics with the results obtained by a simulation that uses $k - \epsilon$ RNG model. Moreover, we show the contour plots of the vorticity and the turbulent kinetic energy, along with the lateral velocity.

2. Introduction

Accurately predicting unsteady wind loads on rectangular bluff bodies is a critical challenge in modern wind engineering, considering their wide use in building industries. The aerodynamic behavior around these structures is highly complex, governed by shear layer separation, flow reattachment, and vortex shedding. Computational Wind Engineering (CWE) provides a valuable tool to investigate these phenomena, though its accuracy heavily depends on the chosen turbulence modeling. Following the reference study by Ying, Xu and Zhang [1], the main goal of this project is to simulate incompressible flow around a rectangular cylinder to compare the performance of the $k - \epsilon$ RNG and $k - \omega$ standard model. By analyzing key aerodynamic parameters, such as the drag coefficient, lift fluctuations. The aim is to evaluate the physical accuracy of each model in capturing boundary layer separation and wake dynamics.

3. Computational Model and Boundary Conditions

This study investigates an unsteady, two-dimensional flow around a square cylinder, characterized by a reference dimension B . To replicate a subcritical flow regime corresponding to a Reynolds number of $Re \approx 2.2 \times 10^4$, the fluid properties are defined within the solver: the density ρ is set to 1 kg/m^3 (air density) and the dynamic viscosity ν is set to $4.545 \times 10^{-5} \text{ kg/(m} \cdot \text{s)}$ and together with the inlet velocity U_∞ , they represent the reference quantities. The model is, in fact, adimensional with respect to these values. In this simulation, the cylinder is assumed to be fixed with a smooth edge (rugosity is set to 0); a no slip condition is set on the edge itself: $u = v = 0$. A velocity-inlet condition is applied at the upstream edge, where the turbulent state is initialized with an

intensity of 2% and a turbulent length scale of 0.5 m (equivalent to $0.5B$). On the outlet edge, a Dirichlet boundary condition is set on the pressure $p_{rel} = 0$. The upper and lower domain limits are constrained using periodic conditions. A uniform initial condition is applied across the entire computational domain, meaning that all flow variables and turbulence quantities in the internal field are perfectly matched to the values imposed at the inlet boundary initially.

4. Spatial and Temporal Discretization

Regarding the spatial discretization, the computational domain is partitioned into two distinct macro-zones: an outer area (referred to as the EUL region) and an inner area surrounding the bluff body (the ALE region). The outer EUL region is discretized using a mapped face meshing technique, which yields a structured grid composed entirely of quadrilateral cells. This approach efficiently handles the free-stream flow while minimizing the overall computational burden. Conversely, for the inner ALE, an unstructured mesh made of triangular cells is used, the flexibility needed to capture the complex flow behavior near the obstacle. Furthermore, to correctly simulate the boundary layer and the flow separation, a finer layer of quadrilateral cells (grid boundary layer) is placed directly around the solid walls of the cylinder.

The simulation is initially run for 10,000 time steps, so that the flow can reach its developed state. After that, we analyze the lift coefficient C_l to retrieve the approximate number of time steps to bring the simulation at the start of a period ($C_l \simeq 0$). At last, we let the simulation continue for approximately 360 time steps in order to simulate only a period of the periodic state. In all the above procedures we used an advancing time step value of $0.02s$.

5. Turbulence Modeling

The primary goal of this simulation is to understand the differences in behavior between the $k - \epsilon$ RNG and $k - \omega$ turbulence models when applied to a sharp-edged square cylinder. Both models are based on the RANS approach. Instead of calculating every single eddy (they are, in fact, non-scale resolving models), RANS models use the Boussinesq hypothesis to estimate a "turbulent viscosity" that represents the average effect of the turbulence on the main flow. Once the Boussinesq hypothesis is applied to the equations, the problem obtained is not closed: it lacks the equations for the evolution of both the turbulent kinetic energy (k) and its dissipation rate (ϵ). The STD $k - \epsilon$ model calculates these two quantities by solving a transport equation for k , which is derived exactly from the governing physical laws, alongside an equation for ϵ , which relies heavily on empirical parameters. While it works well for streamlined shapes, it has major problems with bluff bodies. Specifically, it produces way too much turbulent energy near the front wall of the cylinder and underestimates Re . To mitigate this issue, the RNG $k - \epsilon$ variant replaces the standard fixed constants with analytically derived coefficients. It also modifies the production of turbulent kinetic energy by making it dependent on the flow's vorticity, effectively

preventing the generation of unrealistically high turbulence levels in stagnation regions. In order to try to fix these near-wall errors, the $k - \omega$ model uses the specific dissipation rate (ω) instead of ϵ . The main advantage of this change is that the mathematical equations do not become unstable near the solid walls, even when the turbulent energy (k) is very small.

6. Results

After running the simulation, we are now ready to show the results. We can look at the contour plots of both vorticity and k .

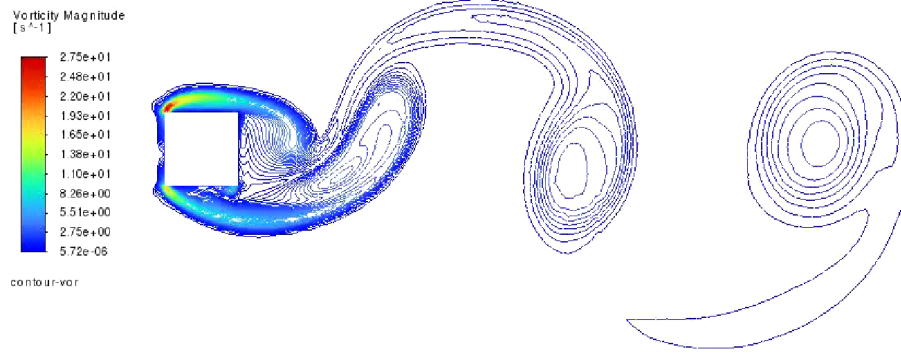


Figure 1: Snapshot of vorticity contour plot.

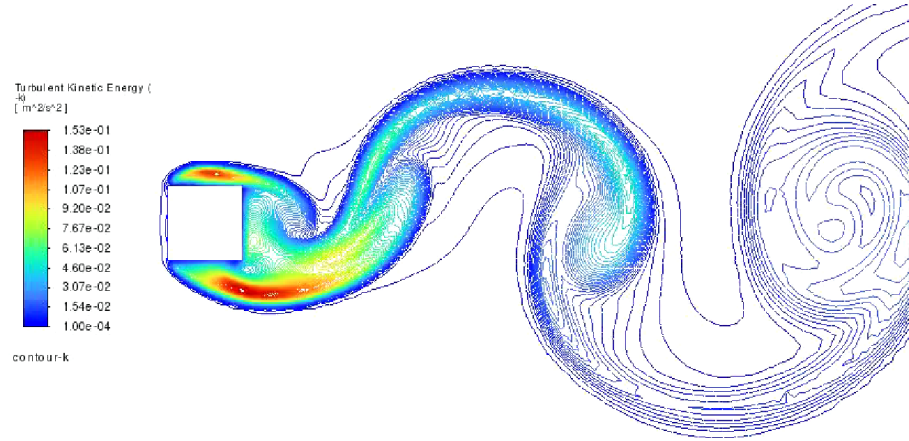


Figure 2: Snapshot of k contour plot.

In figure (1), we can see that the simulation has reached his full development, because there are two vortices behind the bluff body. In the complete video

clip ([2]) we can see that there is more vorticity on the front of the bluff body compared to the $k - \epsilon$ RNG model; this is due to the over prediction of this quantity in our model. In general, the numerical range for the values of the vorticity and k is coherent with the bibliographic data. For a single period we show the values of C_l and C_d .

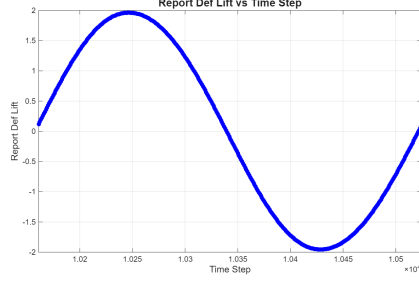


Figure 3: C_l values.

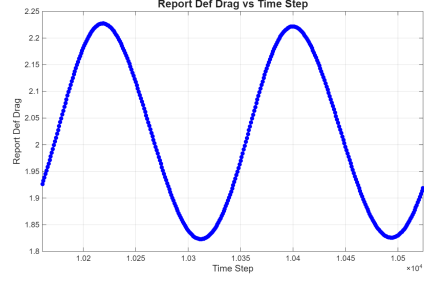


Figure 4: C_d values.

As expected, figure (3) shows that C_l has (almost) zero as starting and ending value. Figure (4) shows that C_d has some oscillations. We can also take a look at the mean value and the rms of the C_p , both evaluated on the center line, i.e. $y = 0$, and compare our results with the ones from the $k - \epsilon$ RNG model; named "DIMAT" in the figures.

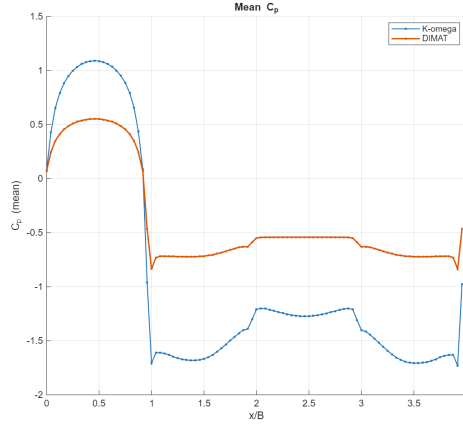


Figure 5: Comparison of mean C_p values along the center line.

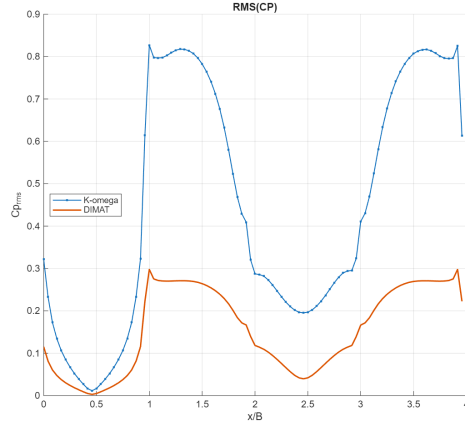


Figure 6: Comparison of rms C_p values along the center line.

In figure (5) we can see that our model overestimates C_p , while not exceeding $C_p = 1$ of a relevant quantity, in the frontal zone, while underestimates it once the flow has separated. Regarding the rms of C_p , figure (6) shows that our model overestimates this quantity throughout all the values of x/B . This happens probably due to the fact that the $k-\omega$ model lacks of the optimizations that are applied in the $k-\epsilon$ RNG model. As our last result, we plot the value of the mean velocity along the line $x = 1$, for all phases.

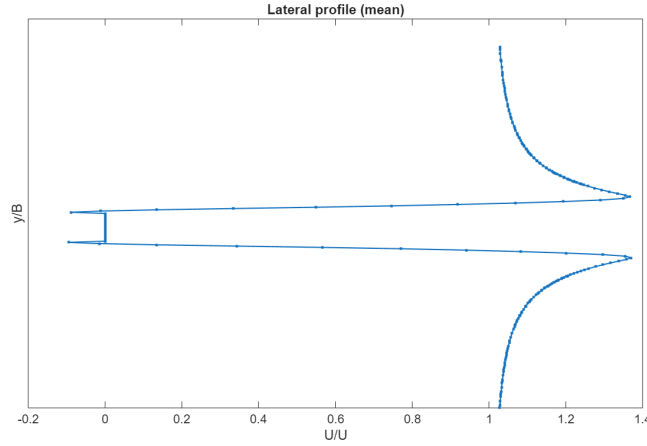


Figure 7: Mean values of u along $x = 1$.

From figure (7) we can see that the profile matches the one presented in the literature for single phases, with some small variations in the numerical values.

7. Conclusions

We can affirm that there are no major differences between the two models in terms of values. By looking at the contour plots, the $k - \epsilon$ RNG model seems to have a slightly better performance. After displaying and comparing the results of the two models we can affirm that the $k - \omega$ standard model overestimates rms of C_p and under estimates its mean in the rear part of the cylinder.

Remark

All the shown results along with the scripts used to retrieve them and the bibliographic data are available at [2].

References

- [1] X. Ying, F. Xu, Z. Zhang, Numerical simulation and visualization of flow around rectangular bluff bodies, in: The Seventh International Colloquium on Bluff Body Aerodynamics and Applications (BBAA7) Shanghai, China, 2012, pp. 272–281.
- [2] S. D. M. Giovanna Maccarone, Github Repository.